

Comment on “Emergence of constructor-based irreversibility in quantum systems: Theory and experiment”

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(Dated: October 2, 2026)

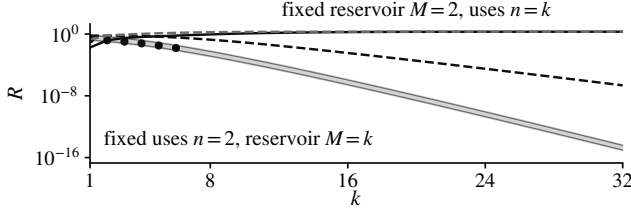


FIG. 1. Current-output infidelity divided by reservoir fidelity at $\eta = \pi/4$, with $p = |0\rangle\langle 0|$ and $\tau = I/2$: solid, $p \rightarrow \tau$; dashed, $\tau \rightarrow p$. The upper pair tends to 2; the falling pair tends to zero. Rational channel updates and certified spectral intervals give the curves and markers. The gray band encloses the forward falling ratio using $e \leq R \leq 4e$, with certified current infidelity e .

We examine the repeated homogenizer of Ref. [1] with all correlations retained. The current-error limits agree in both directions; the published capability infimum fails separately.

The definitions changed between versions. Preprint v1 [2], p. 2, uses a supremum and a two-variable limit; its p. 3 limits take uses first. Published Eqs. (3)–(7), (12)–(13) [1] instead use an infimum, a fixed first-use numerator, and uses before accuracy. The supplement, Eqs. (9)–(10), substitutes the designated initializer and assumes $S(n) \approx S(1)^n$. Reference [3], Eqs. (8)–(13), uses current infidelity; its preprint leaves the joint limit unspecified, while the published Eq. (11) and the thesis [4], Eq. (5.4), take reservoir size first.

Let $U_\eta = \cos \eta I + i \sin \eta S$, $0 < \eta \leq \pi/2$, with inputs a and M cells initially in $b^{\otimes M}$. Write $\rho_{M,n}$ for output n , $\Omega_{M,n}$ for the joint outputs, and $\sigma_{M,n}$ for the reservoir. A fresh b ancilla sweeping k cells defines $\Phi_{b,k}$. Disjoint-gate exchange [5] gives

$$\Omega_{M,n} = \Phi_{b,n}^M(a^{\otimes n}), \quad \sigma_{M,n} = \Phi_{a,M}^n(b^{\otimes M}). \quad (1)$$

Every finite block relaxes to its boundary product state. For a pure boundary, its channel is triangular in excitation sectors, with diagonal moduli $\cos^{r+t} \eta$ and only the vacuum peripheral. Full swap shifts the block. For a mixed boundary, an all-pure-eigenstate branch of a sufficiently long channel power gives strict contraction.

Use squared Uhlmann fidelity, $0 < f = F(a, b) < 1$, and $R_{M,n} = [1 - F(\rho_{M,n}, b)] / F(\sigma_{M,n}, b^{\otimes M})$. Comparing with the invariant grid $b^{\otimes(n+M)}$ gives reservoir fidelity at least f^n . Relaxation therefore yields

$$\lim_{M \rightarrow \infty} R_{M,n} = 0, \quad \lim_{n \rightarrow \infty} R_{M,n} = (1 - f)f^{-M}. \quad (2)$$

Thus $\lim_n \lim_M R = 0$ and $\lim_M \lim_n R = +\infty$; diagonal sequences exclude a joint limit. Fidelity symmetry gives identical limits for the transpose task (Fig. 1).

For the published infimum, set $p = |0\rangle\langle 0|$, $\tau = I/2$, $q = \cos^2 \eta \in (0, 1)$, $M \geq 2$, and $e_M = (1 - \sqrt{1 - q^{2M}})/2$. With first-output channel $\mathcal{O}_p$, the definitions are

$$V_e = \{\omega : F(\mathcal{O}_p(\omega), \tau) \geq 1 - e\},$$

$$S_e(n) = \inf_{\omega \in V_e} F(\omega, \Phi_{p,M}^n(\omega)). \quad (3)$$

For $q \geq 1/3$, choose $\omega = |1\rangle\langle 1| \otimes \tau^{\otimes(M-1)}$. Its first-output Bloch coordinate is $(2q - 1)q^{M-1}$, whose magnitude is at most q^M , so $\omega \in V_{e_M}$. For $q < 1/3$, keep the first cell in $|1\rangle$, give the second coordinate $z_2 = q(1 - 2q)/(1 - q) \in [0, 1]$, and put the rest in τ . The second update $z \mapsto qz + (1 - q)z_2$, starting at $z = 2q - 1$, gives zero: the first output is exactly τ .

Both product initializers are orthogonal to the attractor $p^{\otimes M}$. Hence $S_{e_M}(n) \rightarrow 0$ and $e_M/S_{e_M}(n) \rightarrow +\infty$, contradicting the published forward uses-first conclusion at every strict partial swap. A surviving asymmetry under whole-memory return is analyzed in Ref. [6].

The author selected the research questions and is responsible for the content. GPT-6 Astra (OpenAI) assisted with derivations, source comparison, and writing, and Claude (Anthropic) was used for review and editing.

Data availability. Exact plotting data and code are available at github.com/SamMausberg/returning-constructor.

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